Supplementary Material: A Multidimensional Agent-Based Modeling and Simulation Framework for Urban Water Demand Forecast

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Abstract

A long-term forecast of urban water demand requires modeling the complex interplay of environmental, technological, and socioeconomic factors. However, traditional aggregated models often fail to capture spatio-temporal heterogeneity and uncertainty in human behavioral dynamics. In this paper, we propose a multidimensional framework, the Agent-Based Model for Water Prediction (ABM-WP), to simulate high-resolution neighborhood-scale water consumption.

Designed using the Tropos methodology and implemented on the GAMA platform, the ABM-WP framework utilizes specialized Analyzer, Predictor, and Communicator agents to process georeferenced household data. The system explicitly maps local observations to three simulated behavioral profiles (environmentalist, moderate, and wasteful) and quantifies the impact of demographic growth and dynamic income elasticity on water demand. To evaluate its predictive stability and ability to capture socioeconomic tipping points, ABM-WP is benchmarked against statistical forecast models, such as Seasonal Autoregressive Integrated Moving Average (SARIMA), and machine learning algorithms, including Long Short-Term Memory (LSTM), Multilayer Perceptron (MLP), and Support Vector Regression (SVR). Using an extensive historical dataset from Salvador, Brazil, the model forecasts 18 distinct scenarios for 2026 to 2035. Statistical validation using Analysis of Variance (ANOVA) and Tukey tests demonstrates that the framework identifies critical divergences from historical inertia, with the environmentalist scenario projecting significant consumption reductions of 48,864 m^3/year. Results show that ABM-WP outperforms purely autoregressive approaches by capturing the non-linear emergence of behavioral and income transitions, providing a robust decision-support tool for urban water sustainability.

Keywords: LSTM, SARIMA, SVR, Agent-Based Model, Urban Water Sustainability

1 Introduction

This supplementary material provides documentation of the Agent-Based Model for Water Prediction (ABM-WP). Section 2 presents an overview of water demand forecasting frameworks with ABM-WP. Section 3 presents the description of the Performance Measures, Environment, Actuators, and Sensors (PEAS) for the three agents of ABM-WP. Section 4 details the project design developed using the Tropos methodology. Section 6 characterizes the consumption profiles by sector in the Itapuã neighborhood. Section 7 explore combined input variables in benchmark forecasting models. Section 8 show the algorithm for coordinate transformation. Section 9 presents the comparative results of the 18 ABM-WP scenarios in relation to the SARIMA model. Section 10 displays the monthly water consumption projections for all scenarios, along with the LSTM and SARIMA models. Finally, Section 11 presents details of the Shapiro-Wilk normality test results.

2 Overview of Water Demand Forecasting Frameworks

Table 1 presents an overview of water demand forecasting frameworks organized by reference, approach, dataset location and scale, primary contribution, and validation techniques compared to ABM-WP.

Table 1: Overview focusing on ABM, categorized by solution approach, dataset location and scale, primary contribution, and validation technique. Source: Elaborated by the author.

Reference	Approach	Dataset (location & scale)	Primary Contribution	Validation technique
Galán et al. [1]	ABM + GIS	Valladolid, Spain (2,000 agents)	Integration of spatial and social dynamics for domestic water management.	Statistics, Time Series
Koutiva and Makropoulos [2]	ABMS	Athens, Greece (3,000 households)	Evaluation of the efficiency of management policies during drought periods.	Time Series
Sharvelle et al. [3]	IUWM (GIS-based)	Fort Collins, USA (Block groups)	Assessment of demand management under climate and land use change scenarios.	Statistics, Time Series
Darbandsari et al. [4]	ABMS	Tehran, Iran (3,172 households)	Simulation of behavioral characteristics and social interactions for demand management.	Statistics, Time Series
Rasoulkhani et al. [5]	ABMS	Miami Beach, USA (380 households)	Analysis of the influence of income, pricing, and rebate policies on conservation technology adoption.	Statistics, Sensitivity
Searle and Harper [6]	ABMS	Cape Town, South Africa (1,000 households)	Investigation of residential water conservation tendencies in response to demand management strategies.	Statistics, Time Series
Perelló-Moragues et al. [7]	Socio-cognitive ABM	Barcelona, Spain (300 households)	Simulation of complex decision-making via a socio-cognitive architecture for urban demand.	Statistics, Expert Knowledge
Canales et al. [8]	GW-ABM (Review)	Global (77 studies)	Identification of challenges in the simulation of groundwater and society.	–
Vidal-Lamolla et al. [9]	ABMS	Barcelona, Spain (1,200 households)	Simulation of socio-cognitive agents for the assessment of policy implementation.	Expert Knowledge
Botero-Acosta et al. [10]	SWAT (Hydrologic)	Kaskaskia River, USA (Watershed)	Estimation of the isolated impacts of urban expansion on streamflow under climate change.	Statistics, Time Series
Ehsanitabar et al. [11]	ABMS	Tehran, Iran (5,000 households)	Management of reservoir demand via social and hydrological interactions.	Statistics, Expert Knowledge
ABM-WP framework	Spatially explicit ABMS (GAMA)	Salvador, Brazil (19,630 households)	Comparative analysis of ABM against statistical and ML approaches, regarding the influence of demographics and income dynamics.	Statistics, Time Series, Expert Knowledge

3 PEAS Description

The task environment for each of the three agents of ABM-WP includes the Analyzer Agent (AA) in Table 2, the Predictor Agent (PA) in Table 3, and the Communicator Agent (CA) in Table 4.

Table 2: PEAS description for the AA, responsible for data collection, preprocessing, and analysis. Source: Elaborated by the author.

PEAS	Description
Performance	Quantity and quality of data processed per consumer unit
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Analyze family-grouped data
	Compare and process consumption
	Send processed data to predictor agent
Sensors	Residence geolocation
	Residence types and profile
	Population growth

Table 3: PEAS description for the PA, responsible for making scenario simulations. Source: Elaborated by the author.

PEAS	Description
Performance	Maintain prediction quality with minimal Mean Absolute Percentage Error (MAPE)
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Select optimal prediction model
	Generate family consumption forecasts
	Send forecasts to AC
Sensors	Data provided by AA

Table 4: PEAS description for the CA, responsible for consolidating the generated forecasts. Source: Elaborated by the author.

PEAS	Description
Performance	Number of reports with consumption forecasts issued
Environment	Virtual with georeferenced addresses. Classified as Non-deterministic, Dynamic (independent changes), Discrete, Partially observable, Sequential (history-dependent)
Actuators	Send water demand reports
	Distribute to PA water demand reports
Sensors	Data generated by the PA

4 Project Design

The project design of the ABM-WP framework was modeled using the Tropos methodology, illustrated in Figures 1 and 2, which show the three agents responsible for forecasting the water demand and the interaction between human actors (family, operator, meter, attendant, technician, and meter).

Figure 1 presents the project's final requirements, highlighting key Tropos elements (Tasks, Resources, Goals, and Dependencies) to ensure that each agent operates in a coordinated manner to achieve the main objective. The AA collects and processes historical data on consumption, type of residence, and family income, grouping them by consumer unit and preparing them for the next step. The PA makes individualized consumption forecasts, one agent instantiated for each consumer unit. Each agent forecasts household consumption, taking into account its behavior profile and relevant variables. Finally, the CA consolidates the forecasts, compares them with historical data, and generates reports made available for use. This agent-based approach presents a modular, scalable structure, with each agent playing a specific role, ensuring accurate forecasts and providing more detailed information for efficient planning in water resource management.

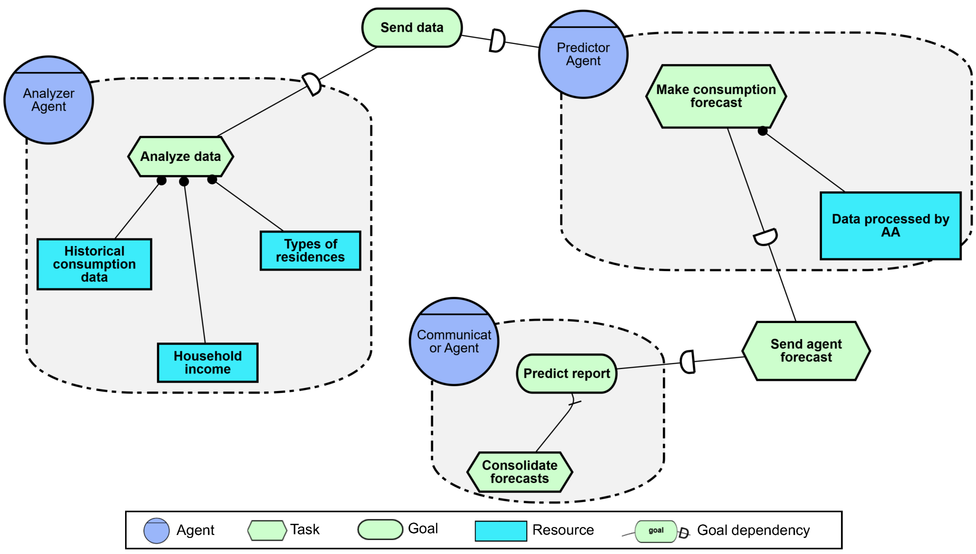

Fig. 1: Tropos final requirements model with the AA, PA, and CA, using the piStar tool. Source: Elaborated by the author.

Figure 2 presents the project architecture, where the Operator is the central point of progression, managing data dependencies with the Family (household data provider) and the Meter (responsible for recording monthly and historical consumption). Operational intelligence is distributed across three specialized agents: (i) the Analyzer Agent, responsible for analyzing, preprocessing, and consolidating customer data; (ii) the Predictor Agent, generating individualized consumption forecasts per family and per scenario; and (iii) the Communicator Agent, which prepares and consolidates scenario predictions. The workflow is driven by dependencies and objectives.

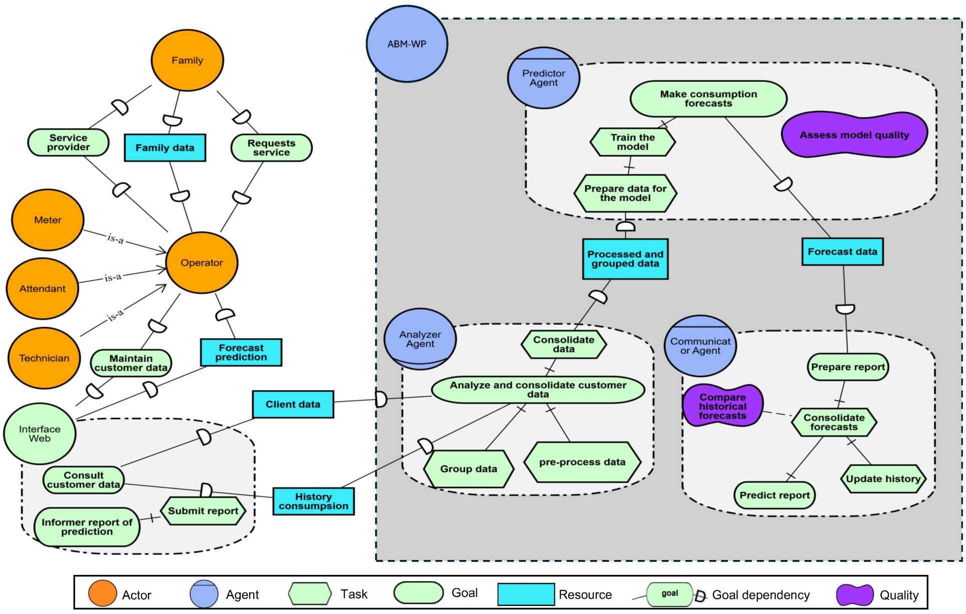

Fig. 2: Architecture of the ABM-WP framework, encompassing the interaction between actors, tasks, and agents.

5 Itapuã Neighborhood Consumption Profiles by Sector

The tables describe 102 sectors using data from April 2026. Each table lists the sector code (CD_SETOR), the number of total consumers (total_consumers), the number of environmentalist consumers (environmentalist_count), the number of moderate consumers (moderate_count), the number of wasteful consumers (wasteful_count), the average daily consumption (avg_daily_consump
tion), the percentage of environmentalist consumers (pct_environmentalist), the percentage of moderate consumers (pct_moderate), and the percentage of wasteful consumers (pct_wasteful).

Figure 3 presents the consumption profiles by sector in the Itapuã neighborhood with the demarcation of the 102 sectors and the location of the Abaeté Metropolitan Park.

Fig. 3: Demarcation of the Itapuã neighborhood, separated by 102 sector limits in yellow, with the Abaeté Metropolitan Park visible on the right in green, located within an environmental preservation area. Active water connections are those represented by blue circles. Source: Elaborated by the author.

6 Itapuã Neighborhood Consumption Profiles by Sector

Tables 5, 6, and 7 describe 102 sectors using data from April 2026. Each table lists the sector code (CD_SETOR), the number of total consumers (total_consumers), the number of environmentalist consumers (environmentalist_count), the number of moderate consumers (moderate_count), the number of wasteful consumers (wasteful_count), the average daily consumption (avg_daily_consumption), the percentage of environmentalist consumers (pct_environmentalist), the percentage of moderate consumers (pct_moderate), and the percentage of wasteful consumers (pct_wasteful).

7 Combined input variables (consumpsion, population and income growth) in benchmark forecasting models

In addition to implementing statistical and machine learning models using only historical consumption time series, other models were also implemented using the three combined input variables (historical consumption, population growth rate, and income growth rate) to verify whether this variation would yield better results. However, the addition of these auxiliary variables did not improve the models' performance, indicating that the superiority observed in the ABM-WP cannot be attributed to differences in the level of input information among the benchmark models, as can be seen in the Table 8.

8 Algorithm for coordinate transformation

Algorithm 1 presents a direct coordinate transformation from geographic (latitude ϕ, longitude λ) coordinates in the WGS84 datum (EPSG:4326) to projected Universal Transverse Mercator (UTM) coordinates (easting E, northing N) [12] in the Sistema de Referencia Geocentrico para las AmericaS - SIRGAS2000 reference system (EPSG:31984, Zone 24S) [13, 14].

Algorithm 1 WGS84 to SIRGAS2000 UTM coordinate transformation.

Require: Latitude ϕ in decimal degrees (WGS84), Longitude λ in decimal degrees (WGS84)
Ensure: Projected coordinates (Easting E, Northing N) in meters (SIRGAS 2000 UTM 24S)

```
Initialize:
coord_transformer ← Transformer.from_crs(source_crs = "EPSG : 4326",
target_crs = "EPSG : 31984",
always_xy = True)
function ConvertGeographicToProjected(φ, λ)
    (E, N) ← coord_transformer.transform(x ← λ, y ← φ)
    return (E, N)
end function
return coord_transformer
```

Table 5: Consumption profiles by sector in the Itapuã neighborhood (Part 1 of 3).

CD_SETOR	Total	Env.	Mod.	Waste.	Avg. Daily	Pct Env.	Pct Mod.	Pct Waste.
S032	317	176	26	115	133.54	55.52	8.20	36.28
S057	310	181	24	105	127.91	58.39	7.74	33.87
S054	304	177	42	85	103.33	58.22	13.82	27.96
S096	290	193	14	83	98.78	66.55	4.83	28.62
S007	284	179	14	91	140.06	63.03	4.93	32.04
S002	280	135	26	119	129.75	48.21	9.29	42.50
S041	278	135	27	116	140.66	48.56	9.71	41.73
S019	276	206	24	46	71.45	74.64	8.70	16.67
S055	261	147	32	82	102.48	56.32	12.26	31.42
S068	242	130	27	85	134.99	53.72	11.16	35.12
S014	235	173	18	44	72.48	73.62	7.66	18.72
S030	233	147	21	65	103.85	63.09	9.01	27.90
S095	233	128	25	80	115.45	54.94	10.73	34.33
S099	231	154	17	60	122.88	66.67	7.36	25.97
S035	226	47	24	155	182.20	20.80	10.62	68.58
S005	225	115	19	91	174.56	51.11	8.44	40.44
S036	225	24	16	185	209.24	10.67	7.11	82.22
S016	222	112	17	93	164.07	50.45	7.66	41.89
S011	219	97	16	106	221.99	44.29	7.31	48.40
S024	217	142	19	56	93.76	65.44	8.76	25.81
S031	212	76	18	118	171.29	35.85	8.49	55.66
S098	211	147	19	45	88.71	69.67	9.00	21.33
S004	209	98	20	91	149.42	46.89	9.57	43.54
S083	208	128	23	57	103.09	61.54	11.06	27.40
S074	208	124	17	67	94.16	59.62	8.17	32.21
S051	205	133	19	53	91.23	64.88	9.27	25.85
S020	199	97	15	87	132.31	48.74	7.54	43.72
S070	198	68	16	114	298.36	34.34	8.08	57.58
S026	197	109	17	71	110.38	55.33	8.63	36.04
S015	195	119	13	63	107.19	61.03	6.67	32.31
S039	191	158	11	22	55.14	82.72	5.76	11.52
S029	186	140	9	37	75.46	75.27	4.84	19.89
S042	186	128	8	50	89.61	68.82	4.30	26.88
S093	184	140	8	36	58.97	76.09	4.35	19.57

Table 6: Consumption profiles by sector in the Itapuã neighborhood (Part 2 of 3).

CD_SETOR	Total	Env.	Mod.	Waste.	Avg. Daily	Pct Env.	Pct Mod.	Pct Waste.
S101	184	90	20	74	135.37	48.91	10.87	40.22
S094	184	80	20	84	173.61	43.48	10.87	45.65
S069	183	62	22	99	260.82	33.88	12.02	54.10
S033	179	146	14	19	50.84	81.56	7.82	10.61
S006	179	74	14	91	203.58	41.34	7.82	50.84
S003	172	89	10	73	156.67	51.74	5.81	42.44
S037	171	107	14	50	99.67	62.57	8.19	29.24
S047	171	88	11	72	174.44	51.46	6.43	42.11
S028	169	116	9	44	84.06	68.64	5.33	26.04
S022	168	104	6	58	118.55	61.90	3.57	34.52
S034	165	127	5	33	69.41	76.97	3.03	20.00
S077	164	81	13	70	148.26	49.39	7.93	42.68
S059	163	112	13	38	88.79	68.71	7.98	23.31
S067	153	103	14	36	81.86	67.32	9.15	23.53
S082	150	93	17	40	102.59	62.00	11.33	26.67
S087	148	100	16	32	82.33	67.57	10.81	21.62
S092	147	116	10	21	54.33	78.91	6.80	14.29
S025	145	62	9	74	175.49	42.76	6.21	51.03
S061	141	103	9	29	69.55	73.05	6.38	20.57
S038	140	63	20	57	131.06	45.00	14.29	40.71
S040	138	101	11	26	76.02	73.19	7.97	18.84
S084	138	59	17	62	133.65	42.75	12.32	44.93
S013	137	52	11	74	374.56	37.96	8.03	54.01
S062	136	119	4	13	38.32	87.50	2.94	9.56
S076	136	64	11	61	148.74	47.06	8.09	44.85
S050	135	86	14	35	97.03	63.70	10.37	25.93
S027	134	73	13	48	120.95	54.48	9.70	35.82
S075	132	96	8	28	75.03	72.73	6.06	21.21
S043	132	72	16	44	141.38	54.55	12.12	33.33
S056	131	88	8	35	102.23	67.18	6.11	26.72
S023	129	80	11	38	87.58	62.02	8.53	29.46
S073	128	60	7	61	170.84	46.88	5.47	47.66
S010	123	72	11	40	143.48	58.54	8.94	32.52
S053	117	83	8	26	72.16	70.94	6.84	22.22

Table 7: Consumption profiles by sector in the Itapuã neighborhood (Part 3 of 3).

CD_SETOR	Total	Env.	Mod.	Waste.	Avg. Daily	Pct Env.	Pct Mod.	Pct Waste.
S086	115	88	7	20	68.44	76.52	6.09	17.39
S046	114	64	10	40	122.09	56.14	8.77	35.09
S071	113	46	12	55	153.88	40.71	10.62	48.67
S008	112	69	8	35	109.21	61.61	7.14	31.25
S049	111	86	5	20	66.24	77.48	4.50	18.02
S102	109	24	10	75	411.99	22.02	9.17	68.81
S012	106	27	7	72	285.91	25.47	6.60	67.92
S009	104	43	4	57	173.22	41.35	3.85	54.81
S103	104	20	13	71	310.81	19.23	12.50	68.27
S078	101	81	9	11	52.55	80.20	8.91	10.89
S088	96	54	10	32	132.94	56.25	10.42	33.33
S080	95	74	8	13	66.40	77.89	8.42	13.68
S052	93	19	4	70	494.99	20.43	4.30	75.27
S100	90	64	7	19	329.13	71.11	7.78	21.11
S045	88	42	7	39	191.87	47.73	7.95	44.32
S064	88	35	8	45	176.48	39.77	9.09	51.14
S072	85	45	5	35	308.25	52.94	5.88	41.18
S090	84	55	5	24	109.39	65.48	5.95	28.57
S058	84	42	6	36	156.80	50.00	7.14	42.86
S018	81	40	8	33	230.52	49.38	9.88	40.74
S048	80	30	9	41	217.21	37.50	11.25	51.25
S065	79	34	10	35	142.43	43.04	12.66	44.30
S079	75	49	9	17	84.13	65.33	12.00	22.67
S044	72	39	4	29	121.00	54.17	5.56	40.28
S060	61	51	2	8	42.86	83.61	3.28	13.11
S021	61	45	3	13	93.59	73.77	4.92	21.31
S063	55	51	1	3	28.45	92.73	1.82	5.45
S066	50	35	5	10	78.54	70.00	10.00	20.00
S089	45	25	5	15	113.03	55.56	11.11	33.33
S091	26	12	1	13	144.22	46.15	3.85	50.00
S085	2	1	0	1	37217.21	50.00	0.00	50.00
S097	1	1	0	0	99.27	100.00	0.00	0.00

Table 8: Percentage of MAPE obtained on the test dataset for each machine learning algorithm, including model name, percentage with a single input variable, best configuration found, percentage with three input variables, and best configuration found.

Model	**MAPE (%)**	**Configuration**	**MAPE Multi (%)**	**Configuration Multi**
LSTM	0.04	neurons = 64, dropout = 0.2, dense = (25, 1), learning_rate = 0.01	0.04	neurons = 256, dropout = 0.2, dense = (25, 1), learning_rate = 0.01
MLP	0.06	layers = (100, 50, 50, 25, 1)	0.07	layers = (100, 50, 50, 25, 1)
SVR	0.05	epsilon = 0.01, gamma = scale, kernel = linear	0.10	epsilon = 0.001, gamma = 0.1, kernel = rbf
SARIMA	0.05	order = (9, 0, 1), seasonal_order = (1, 1, 1, 12)	0.07	order = (1, 0, 1), seasonal_order = (1, 1, 1, 12)

9 Comparative Results of ABM-WP Scenarios vs. SARIMA Model

Figure 4 shows the annual discrepancies between each of the 18 ABM scenarios and the SARIMA statistical forecast over the 2026–2035 horizon. Positive values indicate that the ABM scenarios project consumption above the SARIMA baseline, while negative values indicate lower consumption. The scenarios combining population growth with income dynamics (S_IV–S_IX) exhibit the largest positive discrepancies, reaching close to 100,000 m^3 by 2035, with the income imbalance scenarios (S_VI and S_IX) showing the largest deviations. The specific waste scenarios (S_III and S_XIII–S_XV) also exhibit substantial positive discrepancies, though slightly smaller than those in the population-income scenarios. In contrast, the environmentally focused scenarios (S_II and S_X–S_XII) show discrepancies close to zero or slightly negative over the projection period. The random income scenarios (S_XVI–S_XVIII) exhibit intermediate behavior, with discrepancies ranging from 20,000 to 40,000 m^3. These results demonstrate that population growth and income dynamics, and not just behavioral profiles, are primarily responsible for the divergence of the ABM-WP model from statistical predictions.

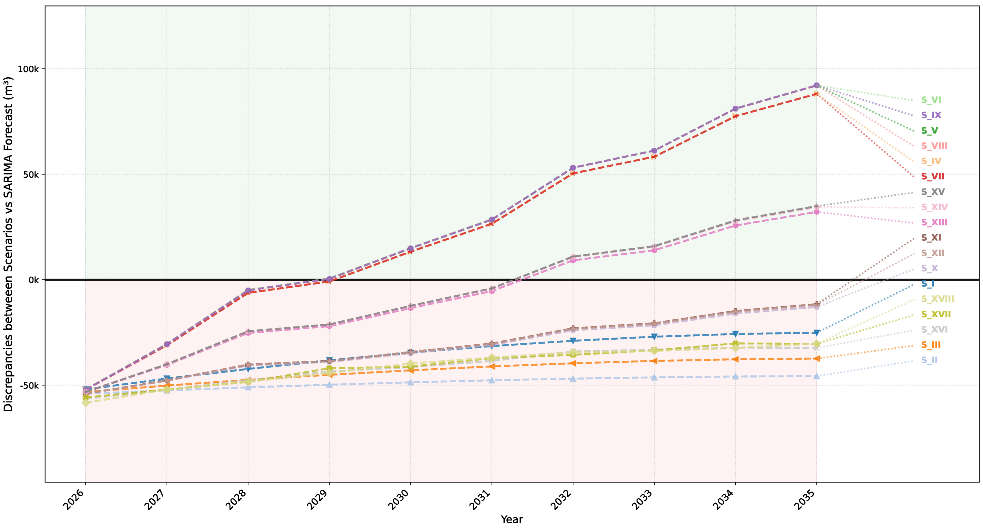

Fig. 4: Annual discrepancies between ABM scenarios and SARIMA forecast (2026–2035). Positive values indicate higher consumption projected by ABM scenarios. Scenarios combining population growth with income dynamics (S_IV–S_IX) show the largest positive discrepancies. Environmentalist scenarios (S_II, S_X–S_XII) show near-zero deviations.

10 Projection of ABM-WP Scenarios vs LSTM and SARIMA Models

Figure 5 presents monthly water consumption projections for all 18 ABM scenarios, alongside LSTM neural network and SARIMA statistical forecasts over the 2026–2035 horizon. While SARIMA and LSTM offer precision in capturing recurring seasonal patterns, ABM-WP offers a strategic advantage: even without fine-tuning for seasonality, it identifies demand trajectories driven by income inequality and behavioral change, something essential for long-term planning in socioeconomic transition scenarios.

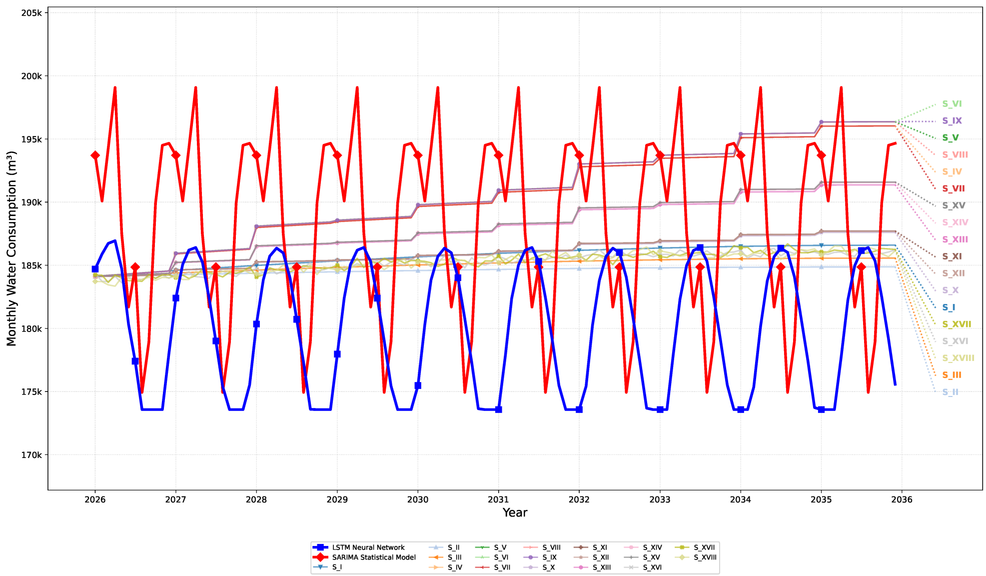

Fig. 5: Monthly water consumption projections (2026–2035) for 18 ABM scenarios, LSTM neural network, and SARIMA statistical model.

11 Shapiro-Wilk Normality Test

Table 9 confirms the normal distribution for all ABM-WP scenarios (p-value > 0.05), unlike the SARIMA model, which rejected normality (p-value $= 0.0000$).

Table 9: Shapiro-Wilk normality test results for water demand scenarios showing the test statistic (W), p-values, and normality assessment for each consumption scenario. Source: Elaborated by the author.

Scenario	**Statistic (W)**	***p*-value**	**Normal Distribution**
S_I	0.9173	0.3347	Yes
S_II	0.9173	0.3352	Yes
S_III	0.9172	0.3343	Yes
S_IV	0.9701	0.8914	Yes
S_V	0.9702	0.8927	Yes
S_VI	0.9703	0.8932	Yes
S_VII	0.9701	0.8916	Yes
S_VIII	0.9702	0.8928	Yes
S_IX	0.9703	0.8933	Yes
S_X	0.9700	0.8908	Yes
S_XI	0.9701	0.8920	Yes
S_XII	0.9701	0.8920	Yes
S_XIII	0.9701	0.8917	Yes
S_XIV	0.9702	0.8930	Yes
S_XV	0.9703	0.8937	Yes
S_XVI	0.9158	0.3235	Yes
S_XVII	0.9255	0.4049	Yes
S_XVIII	0.9192	0.3506	Yes
SARIMA	0.5685	0.0000	No

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